

Q1

Block Diagram and Working of a CRO

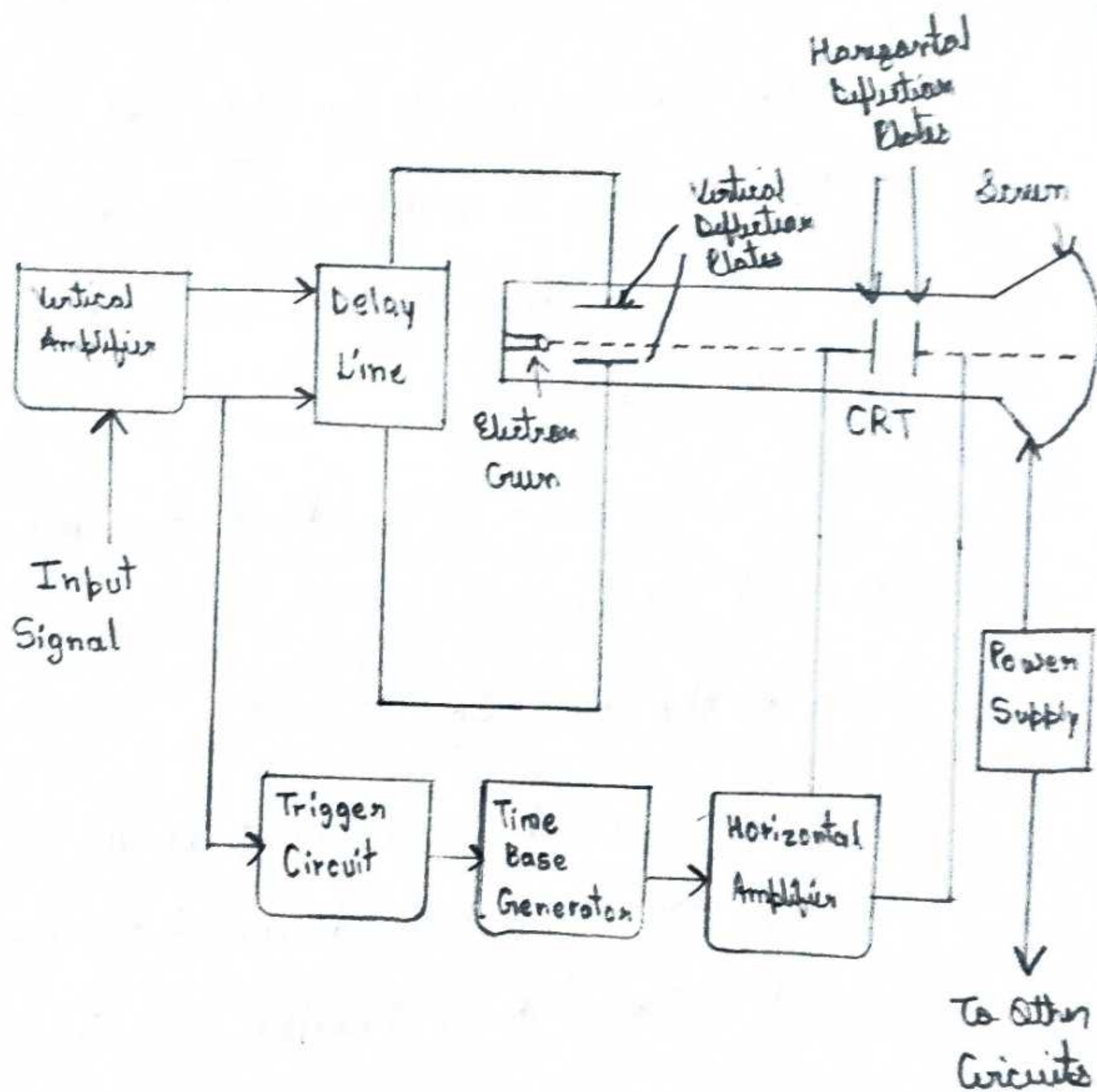
Ans

A Cathode ray Oscillator is a versatile electronic laboratory instrument used to visualize, measure, and analyze electrical signals over time. It essentially plots a two dimensional graph of one or more signals, with voltage on the vertical (Y) axis and time on the horizontal axis.

Here's an Overview of its block diagram and how it works.

Main blocks of a CRO

- 1) Cathode ray Tube - The heart of the Oscilloscope. It generates an electron beam, accelerates it, and deflects it to strike a phosphor-coated screen, creating a visible spot of light.



Block Diagram of a CRO

2) Vertical Amplifier - The input signal is often too weak to deflect the electron beam directly. The Vertical Amplifier increases the signal's amplitude so it can provide enough voltage to the vertical deflection plates.

3) Delay Line - Because the sweep generator (which controls the final horizontal movement) takes a tiny fraction of a second to start, the vertical signal is slightly delayed. This ensures the very beginning (leading edge) of the waveform is visible on the screen.

4) Time Base Generator (Sweep Generator) - This circuit generates a "sawtooth" voltage wave. This saw tooth voltage is applied to the horizontal plates to sweep the electron beam from left to right across the screen at a constant, known speed, creating the time axis.

5) Horizontal Amplifier - Amplifies the smooth wave from the time base generator before applying it to the horizontal deflection plates.

6) Trigger Circuit - If the left to right sweep doesn't start at the exact same point of the input signal's cycle everytime, the wave will look chaotic, moving blur. The Trigger circuit synchronizes the time base generator with the vertical input signal, ensuring the waveform appears perfectly stationary on the screen.

7) Power Supply - Provides two main voltages

- High Voltage - Used to accelerate the electron beam inside the CRT.
- Low Voltage - Used to power the various electronic amplifiers and generate circuits.

Working of a CRO

- 1) Beam Generation - Inside the CRT, an electron gun emits a concentrated, high-speed beam of electrons towards the screen.
- 2) Applying the Input (Y-axis) - The electrical signal you want to measure is fed into the vertical input. It goes through the vertical amplifier and then to the vertical deflection plates inside the CRT. These plates pull the electron beam up and down in exact proportion to the signal's voltage.
- 3) Generating Time (X-axis) - At the same time, the time base generator sends a sawtooth wave to the horizontal deflection plates. This voltage steadily pulls the electron beam from left side of the screen to the right side, then snaps it back instantly to start again.

4) Drawing the wave - Because the beam is being moved horizontally at a steady rate while simultaneously being moved vertically by your input signal, it traces the actual shape of the voltage wave on the glowing phosphor screen.

5) Synchronization - The Trigger circuit ensures that everytime the beam sweeps across the screen, it starts drawing at the exact same voltage level of your input signal. Because this happens dozens or thousands of times a second, your eyes a solid, non flickering graph.

Q2.

Time Base Operation and Specification
of a CRO

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Time Base Operation of a CRO

The CRO is an instrument used to visualize electrical signals, typically displaying voltage against time. The time base generator is the internal circuit responsible for creating the horizontal time axis (x-axis) on the display screen.

•) Sawtooth Waveform Generation - The time base circuit generates a continuous, linearly increasing voltage that abruptly drops back to zero. This creates a sawtooth shaped waveform that is applied directly to the horizontal deflection plates of the CRT.

2) Sweep Phase (Trace) - During the slowly rising, linear portion of the sawtooth wave, the electron beam is smoothly pulled from the left side of the screen to the right at a constant velocity. This constant speed allows the X-axis to accurately represent time.

3) Flyback Phase (Retrace) - Once the beam reaches the right edge, the sawtooth voltage quickly drops rapidly back to its starting value. This quickly snaps the electron beam back to the left side of the screen. During this fraction of a second, a blanking circuit turns off the electron beam so the retrace line does not draw over the displayed signal.

4) Synchronization (Triggering) - To Display a stable, stationary waveform, the time base generator must be synchronised with the incoming vertical signal. The Trigger Circuit ensures that the sawtooth sweep only begins when the input signal reaches a specific voltage threshold, ensuring each trace draws over the exact same path.

Key Specifications of a CRO

When evaluating or operating a CRO, its performance is defined by several critical specifications -

1) Bandwidth - The maximum frequency range the CRO can accurately measure and display without the signal being significantly attenuated. Common analog CRO bandwidths range from 20 MHz to 100 MHz .

2) Rise Time - The time it takes for the Oscilloscope to record a transition from 10% to 90% of a square wave's step pulse. A faster rise time allows the CRO to accurately capture rapidly voltage spikes and high-frequency transients.

3) Time Base Range (Sweep Speed) - The Adjustable range of horizontal sweep speed, usually expressed in time per division on the screen's graticule. This dictates how well the User can expand or compress signals across the screen.

4) Vertical Sensitivity - The ability of the vertical amplifier to display weak signals, expressed in Volts or Milli Volts per Division. Ex - 5 mV/div to 20

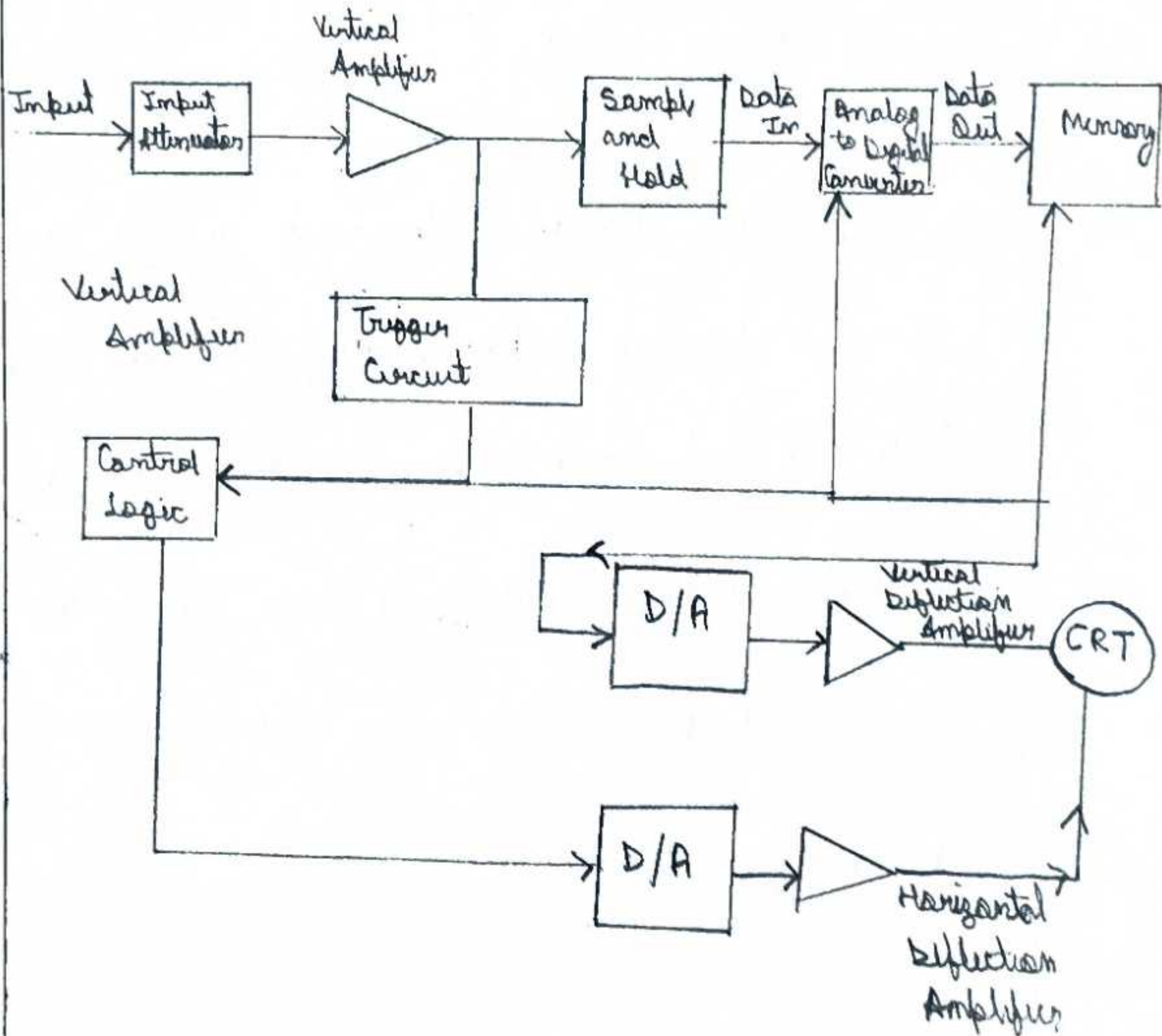
5) Input Impedance - The resistance and capacitance at the input probes. High input impedance is vital to ensure the Oscilloscope does not draw too much current and load down the circuit being tested.

6) Operating Modes - Standard modes include single-channel display, dual channel display (Alternate or chop modes for viewing two signals simultaneously) and X-Y mode where the time base is bypassed and an external signal drives the horizontal direction.

Q3

Block Diagram and Working of a Digital Storage Oscilloscope.

Ans



Block Diagram of a Digital Storage Oscilloscope.

Working of a Digital Storage Oscilloscope -

The Defining feature of a Digital Storage Oscilloscope is that it does not process the continuous analog signal directly to the display ; instead , it translates into a digital dataset first.

Step by step Process -

1) Signal Acquisition and Conditioning

The process begins when the analog signal enters the oscilloscope through a probe.

i.) Attenuation / Amplification - The signal passes through the vertical Amplifier. If the signal is too large, it is attenuated. means it's reduced ; If it's too small, it is amplified.

ii.) Optimizing for the ADC - The goal of this stage is to scale the analog voltage perfectly so that it utilizes the full dynamic range of Analog to Digital Converter.

2) Sampling and Digitization -

The continuous signal then passes to the ADC. Dictated by an internal clock, the ADC takes rapid snapshots of the voltage

sampling and converts each amplitude into a discrete binary number (quantization)

3) Triggering and Storage - The DSO continuously samples and writes these digit values into a circular Memory RAM buffer. When a specific user defined Trigger event occurs like Crossing a set voltage Threshold), the DSO locks in the memory record. This continuous recording allows DSO's to capture and display pre triggered data - The event leads up to the Trigger.

4) Processing and Display - An Internal microprocessor retrieves the stored data, mathematically connects the discrete points to form a continuous curve and can perform automated measurements. Finally

This digital data is mapped to pixels on a display screen, creating a stable visual waveform that can be indefinitely saved or analyzed.

Q4

Digital LCR Bridges.

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A Digital LCR Bridge is an electronic device used to accurately measure the Inductance (L), Capacitance (C) and Resistance (R) of an electronic component.

Unlike traditional analog bridges (like the Wheatstone or Maxwell bridge) that requires a user to manually adjust variable knobs to find a null or balanced point, a digital LCR bridge automates the measurement process using modern digital signal processing and instantly displays the values on a screen.

Key Components include -

1) Internal Oscillator - Generates an AC test signal (usually a sine wave) at selectable frequencies to apply to the component and the current flowing through it.

2) Voltage and Current Measurement Circuits - Analog to Digital Converters measure the voltage drop across the component and the current flowing through it.

3) Phase Detector - Determines the exact phase angle difference between the measured voltage and current.

4) Microprocessors - The brain of the bridge. It takes the digital data from the ADC's and the phase detectors, performs complex impedance calculations, and extracts the L, C and R values.

5) Digital Display - Shows the final measured values, often alongside secondary parameters like the Quality factor or Dissipation Factor.

Working Principle

The Digital LCR bridge does not physically "balance" a bridge circuit. Instead, it measures the complex impedance of the Device under Test based on Ohm's law.

- 1) Stimulus - The Oscillator applies a known AC voltage to the DUT
- 2) Measurement - The Instrument precisely measures the amplitude of the V across DUT, the Amplitude of the current (I) through it and the phase angle between the voltage and Current waveform.
- 3) Impedance Calculation - The microprocessor calculates the Total Impedance magnitude using ratio $\frac{V}{I}$.

4) Parameter Extraction - Using the Total Impedance and the phase angle θ , the microprocessor uses trigonometry to separate the impedance into its real part and Imaginary part.

- If the phase angle shows voltage leading current it calculates Inductance.
- If current leads voltage, it calculates Capacitance.